Do Now · retrieval

Data structure selection challenge

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is an accumulator?

Q2

What is a collision?

Q3

What is a head pointer?

Q4

Which structure is used for Undo and why?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify the operation — The print job that arrived first must leave first.

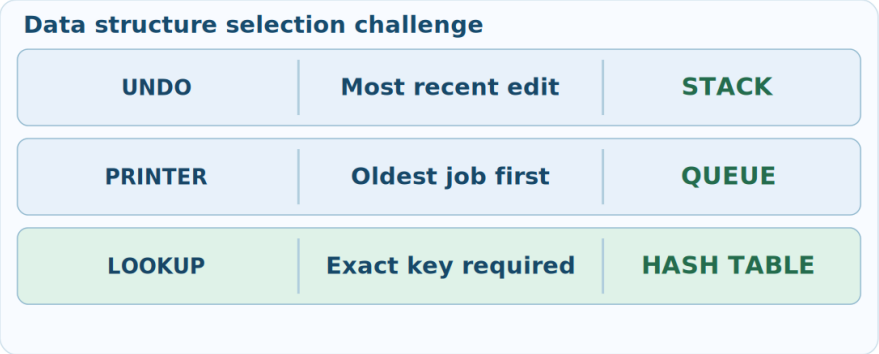
Core idea 2

Match the structure — Enqueue adds jobs; dequeue removes the oldest one. This is a queue.

Core idea 3

Justify and qualify — A stack gives LIFO, which is the wrong order. Other systems may also need priority queues if priority overrides arrival order.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

LIFO — Last In, First Out.

FIFO — First In, First Out.

lookup — Retrieve a value using a key.

balanced BST — Tree kept at relatively small height.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

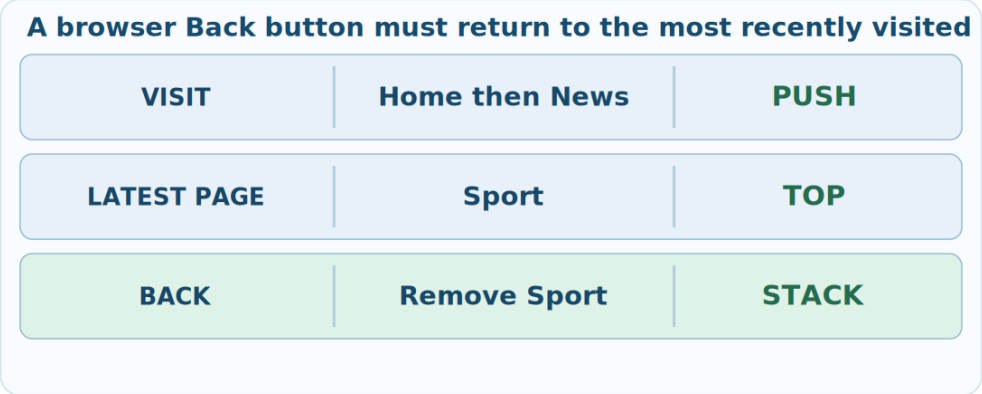
Problem

A browser Back button must return to the most recently visited page. Choose a structure.

Follow the worked method

1. Predict — The last visited page is the first one that must be removed.
2. Explain — Use a stack: push visits and pop the most recent page.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

The first request A must be the first one handled.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Select a structure for processing jobs in arrival order.

Question 2

Choose stack or hash table to retrieve customer records by unique ID. Justify.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare a balanced BST and hash table for lookup and sorted output.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A hash table is always O(1) even in the worst case.

Use key vocabulary from memory

1. Explain LIFO in your own words.

2. Explain FIFO in your own words.

3. Explain lookup in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.